NATIONAL ENGINEERING CENTER
University of the Philippines

1.0

Introduction to Agentic AI

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UP NEC

Module Outline

- Introduction to Agentic AI
  - Capstone Project Case Proposal Generation
- Deep Learning, Transformers and LLMs
  - Capstone Project Business Case Workflow
- Agentic Workflows
  - Case Study and Capstone Project Work
- Agentic RAGs
  - Case Study and Capstone Project Work
- Orchestration
  - Case Study and Capstone Project Work
- Stateful Multi-Agent Systems
  - Case Study and Capstone Project Work
- Evaluation
  - Case Study and Capstone Project Work
- LLMOps
  - Capstone Project Presentation

Outline for this Session

- Introduction to AI
- Agentic AI
- Anatomy of an AI System
- Project Feasibility and Risk Management
- Selecting AI Applications
- Project Proposal Template

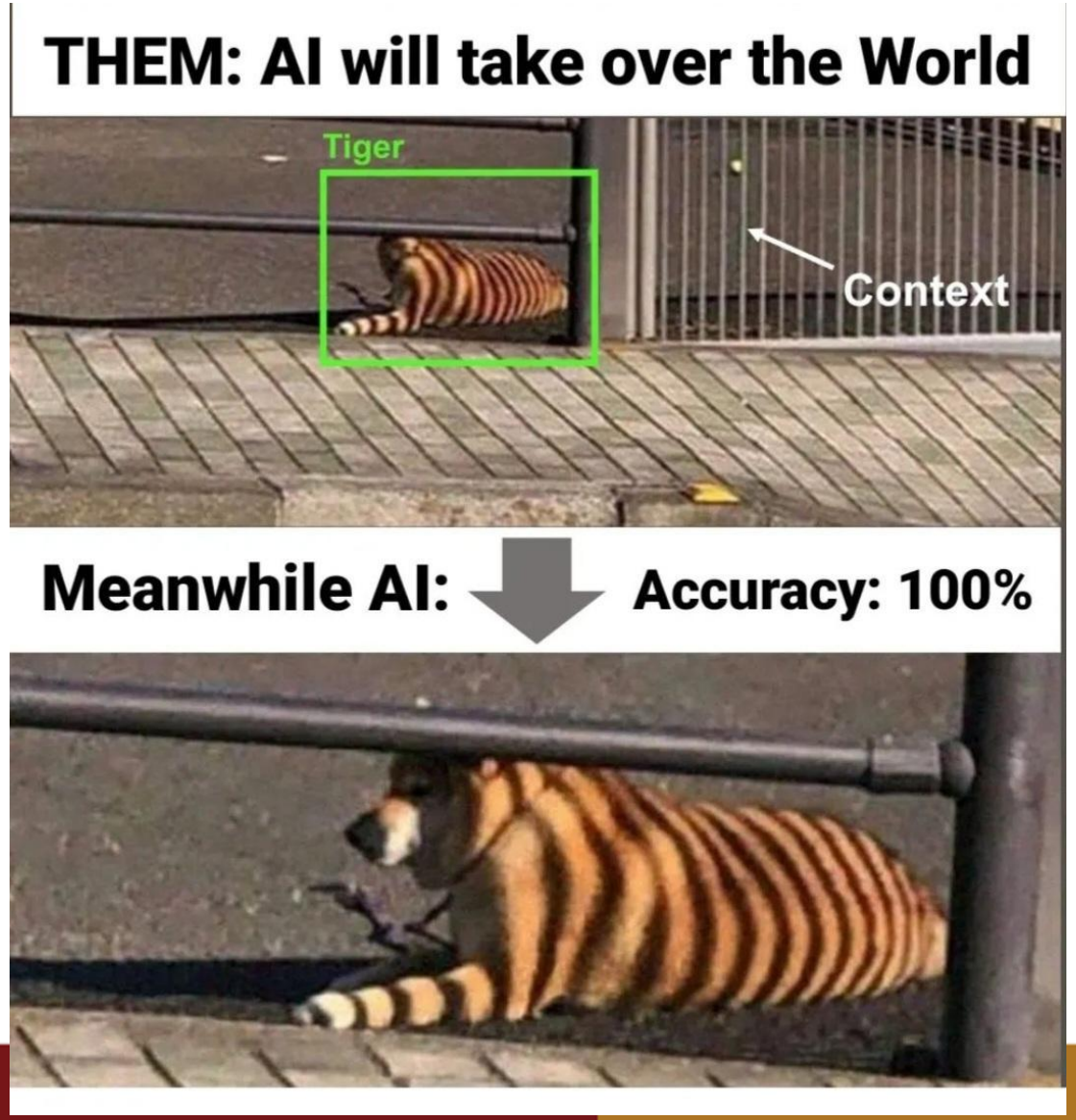
AI

- “Humans should be worried about the threat posed by artificial intelligence.” **~Bill Gates.**



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Also AI:



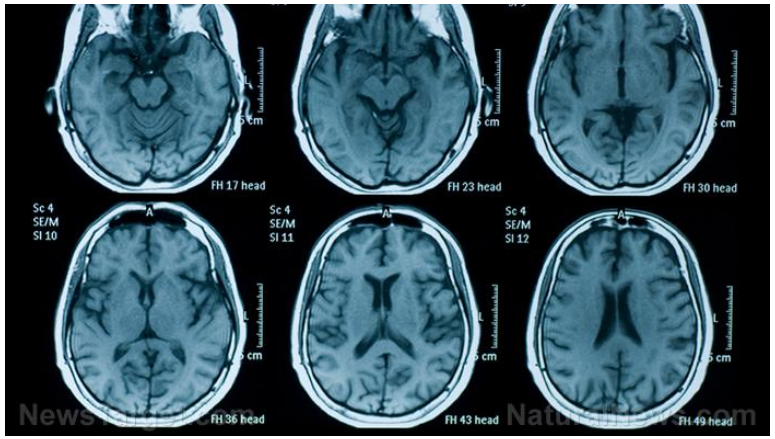
https://www.ebaumsworld.com/pictures/27-fresh-memes-of-the-dankest-kind/87143448/

What is AI?

- UP AI Definition: Artificial Intelligence is the discipline concerned with the design and construction of **intelligent systems** that perceive, reason out, formulate a decision, and act in an environment to achieve a set of measurable goals.



https://www.youtube.com/watch?v=tlThdr3O5Qo



https://www.naturalnews.com/2020-10-01-coronavirus-vaccine-trial-subjects-report-extreme-exhaustion-headaches-cracked-tooth.html

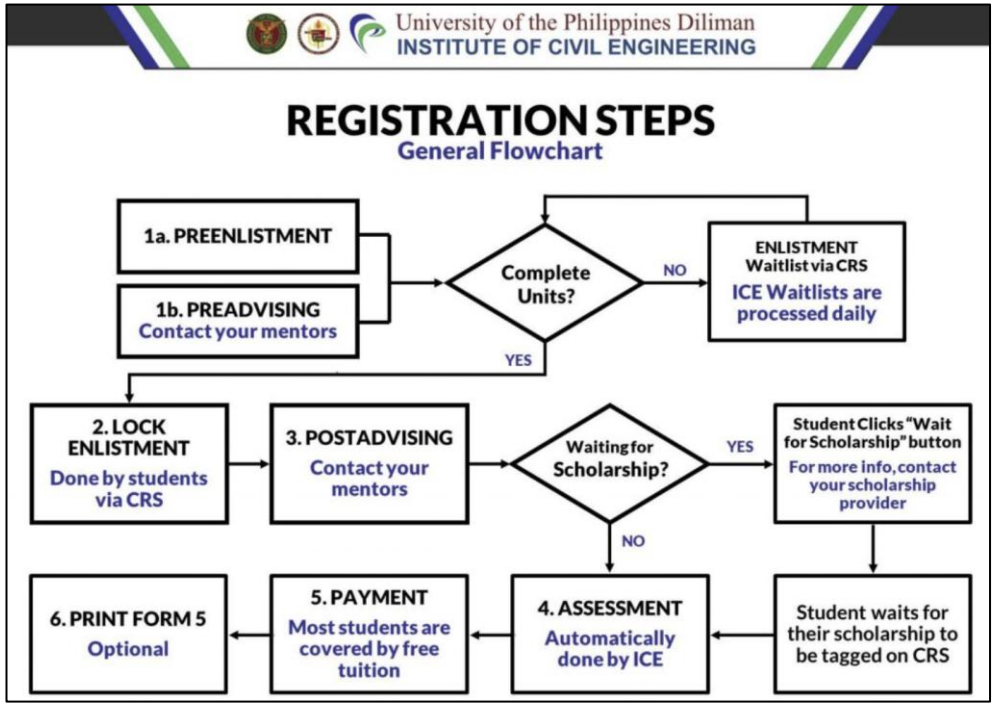
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The Current AI Application Landscape

- Rule-based automation
- Predictive machine learning
- Deep learning
- Generative AI
- Retrieval-augmented generation
- Tool-using AI agents
- Stateful and multi-agent workflows

Traditional Rule-Based Automation

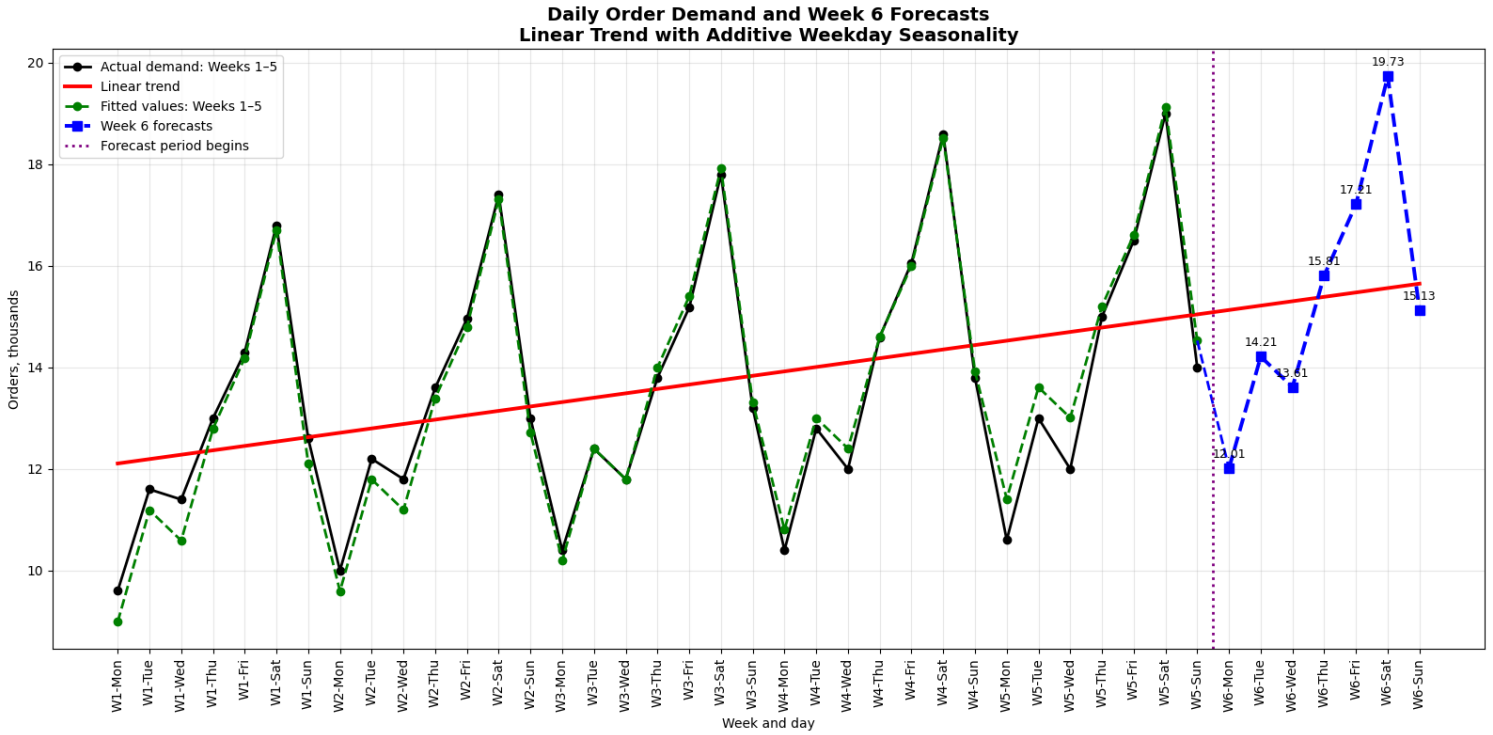
- Traditional **rule-based automation**
  - Uses explicitly programmed conditions
  - Produces predictable outputs
  - Works well for stable and repetitive processes
  - Does not understand natural language semantically
  - Cannot easily handle unfamiliar situations
  - Usually follows a fixed sequence
- Example:
  - if amount > 50000:
  - require_three_quotations = True
  - else:
  - require_three_quotations = False
- Suitable applications:
  - Form validation
  - Tax calculations
  - Eligibility rules



- Workflow routing
- Notifications
- Data transformation

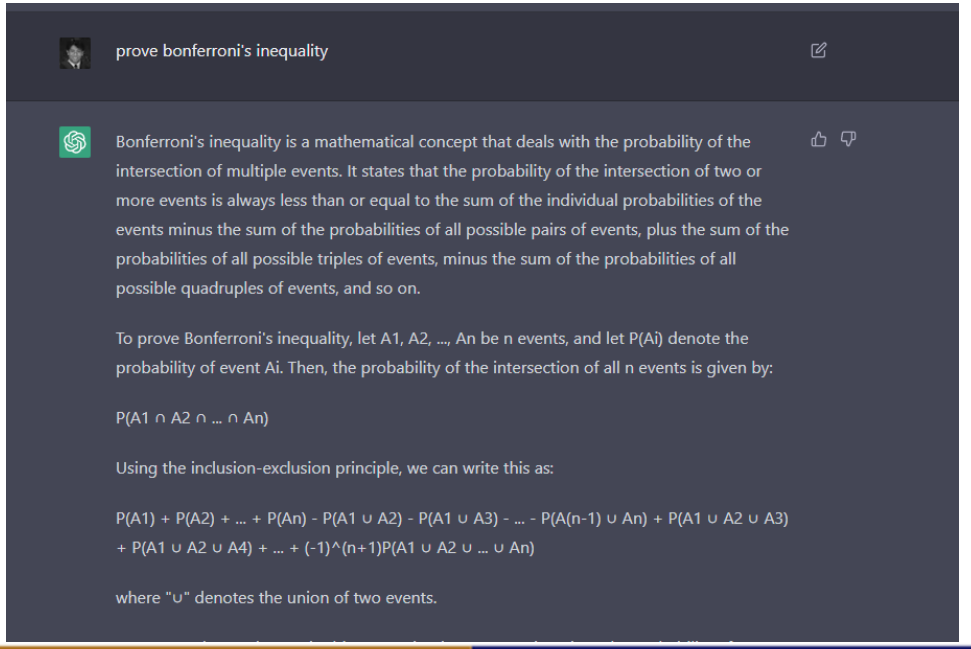
Predictive Machine Learning

- Machine learning is a part of artificial intelligence (AI) that uses data and algorithms to let computers learn and make **predictions** without being explicitly programmed.
- Examples:
  - Customer churn prediction
  - Demand forecasting
  - Fraud detection
  - Credit-risk scoring
  - Equipment-failure prediction



Generative AI

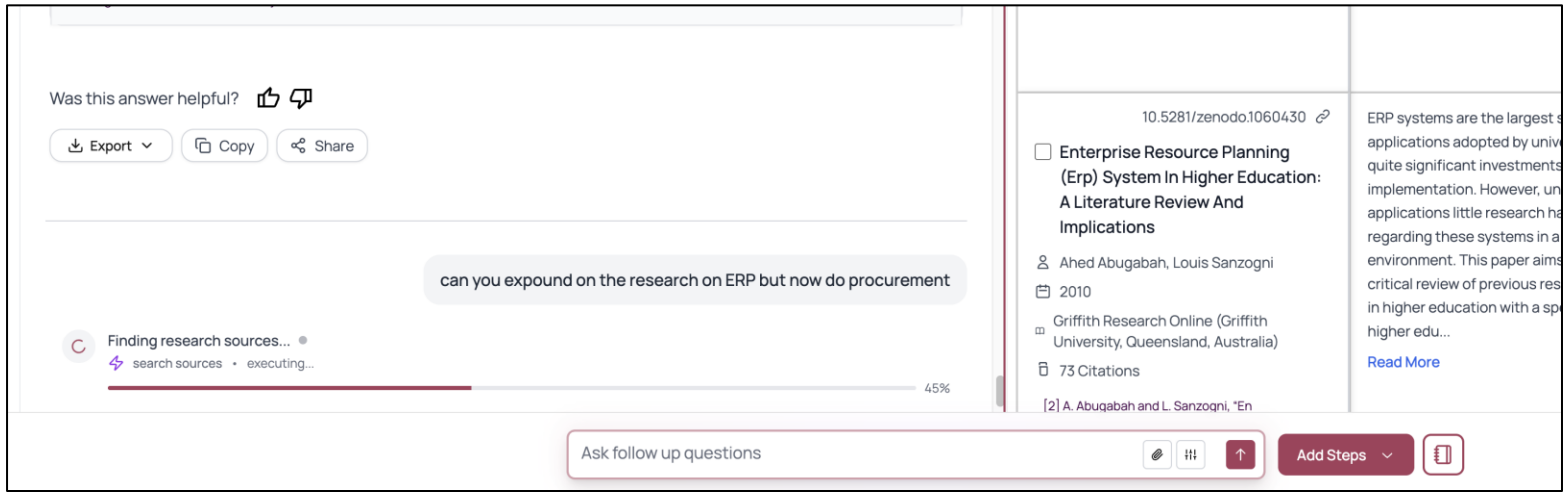
- Generative AI refers to a subfield of machine learning within the rubric of artificial intelligence. GenAI focuses on **creating new content**, such as text, images, videos, code, or synthetic data using machine learning models.
- LLMs are a class of GenAI used to automate the creation of text, code, or data helping in increasing human and/or organizations productivity Examples:
  - Drafting an email
  - Summarizing a report
  - Generating Python code
  - Explaining a policy
  - Creating meeting notes
  - Answering general questions



Retrieval-Augmented Generation

- RAG combines a generative model with **retrieved organizational information**.
- RAG adds:
  - Current organizational documents
  - Internal policies
  - Local procedures
  - Source citations
  - Better factual grounding

ANSWERTHIS



Outline for this Session

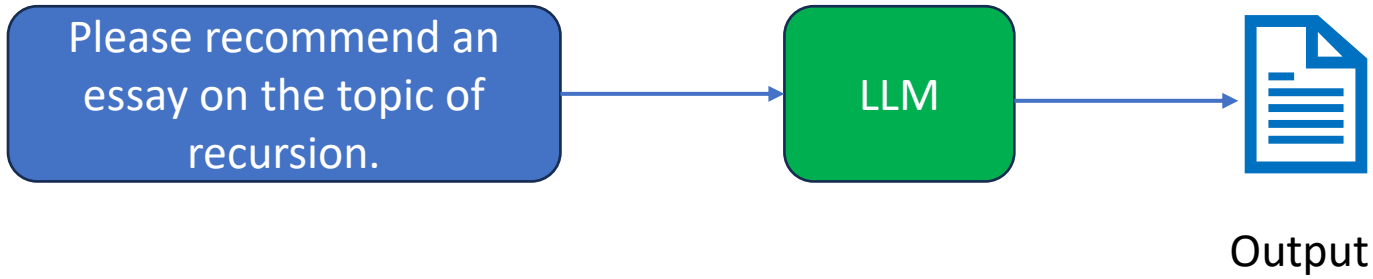
- Introduction to AI
- **Agentic AI**
- Anatomy of an AI System
- Project Feasibility and Risk Management
- Selecting AI Applications
- Project Proposal Template

What is an AI agent?

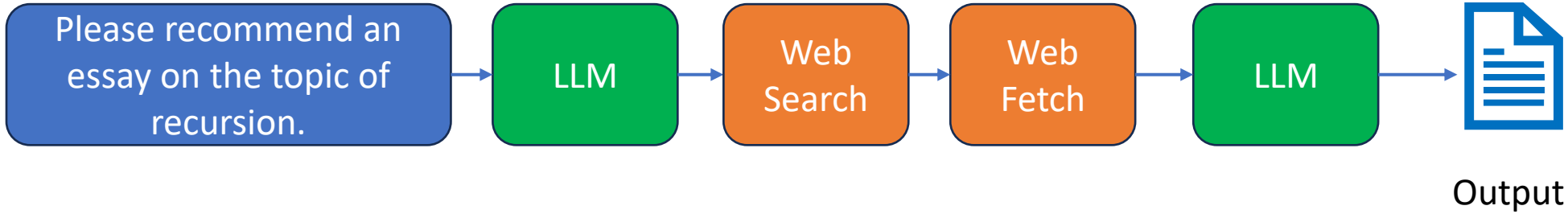
- **Definition: AI Agent**
  - An AI agent is a software system that uses a model to interpret a goal, select among available actions, observe the results, update its state, and continue until it reaches a stopping or escalation condition.
- Agent systems can manage **multiplicity**: Can adapt in real-time to perform the specialized tasks required to bring a process to completion (because these are generative)
- Agent systems can be directed with natural language: You can speak to them naturally enabling technical and non-technical teams to interact
- Agents can work with existing software tools and platforms: These systems are able to integrate with an existing AI ecosystem

Agentic AI

- Non-Agentic Workflow

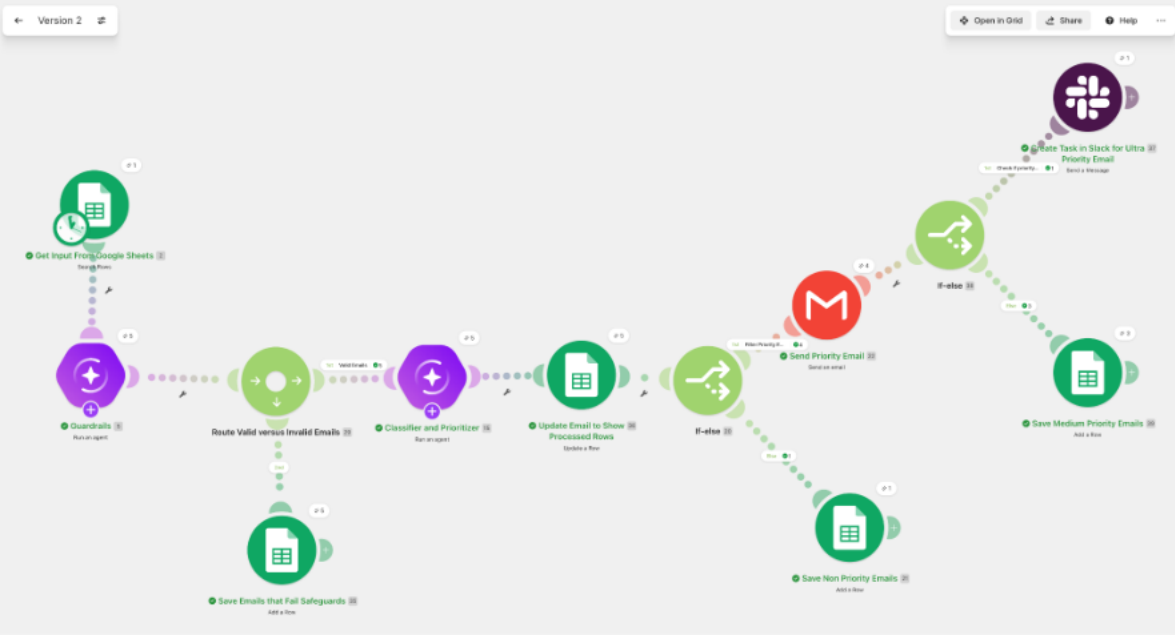


- Agentic Workflow



Agentic AI

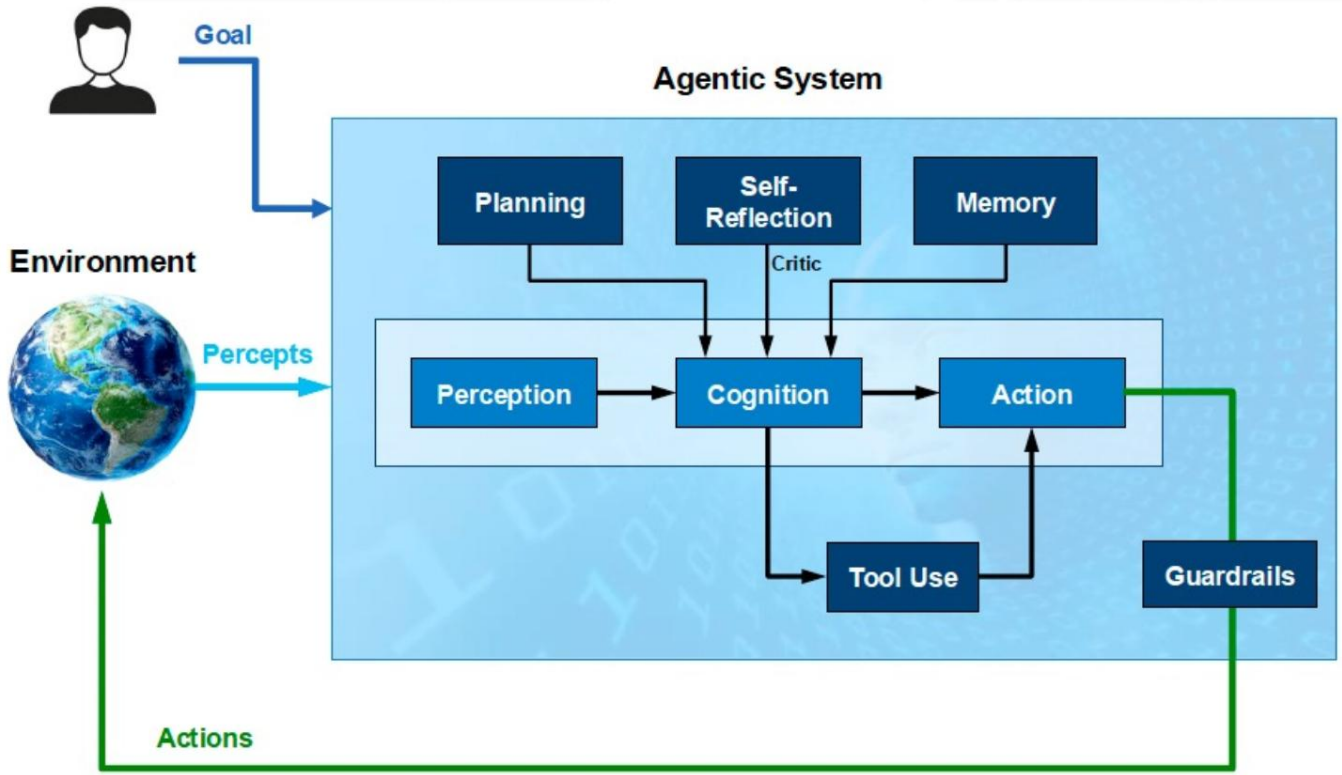
- Agentic AI systems can:
  - Receive a goal
  - Interpret the situation
  - Select from available tools
  - Retrieve information
  - Execute controlled actions
  - Observe results
  - Update state
  - Replan
  - Stop, complete, or escalate



Comparison of major AI application types

Application type	Primary function	Uses tools	Maintains state	Can act
Rule-based automation	Apply fixed rules	Sometimes	Limited	Yes
Predictive model	Predict or classify	No	No	No
Generative AI assistant	Generate content	Usually no	Limited	No
RAG assistant	Answer using documents	Retrieval only	Limited	Usually no
Tool-using agent	Select and execute tools	Yes	Sometimes	Yes
Stateful agent workflow	Coordinate multiple steps	Yes	Yes	Yes
Multi-agent system	Coordinate specialist agents	Yes	Yes	Yes

Key Components of Agentic Systems



Adapted from: Andrew Ng, Founder of DeepLearning.AI and Stanford University Adjunct Professor: https://www.youtube.com/watch?v=sal78ACtGTc

- **Planning:** Break down large tasks into smaller, manageable subgoals, enabling efficient handling of complex tasks
- **Self-reflection and refinement:** The agent can do self-criticism and self-reflection over past actions, learn from mistakes and refine them for future steps, thereby improving the quality of final results
- **Memory:** Short-term memory (context) and long-term memory (world model) to retain and recall information
- **Tool Use:** Agents can employ tools to expand their capabilities
- **Guardrails:** Safe action in the environment

Example: Water Utility Service Response Agent

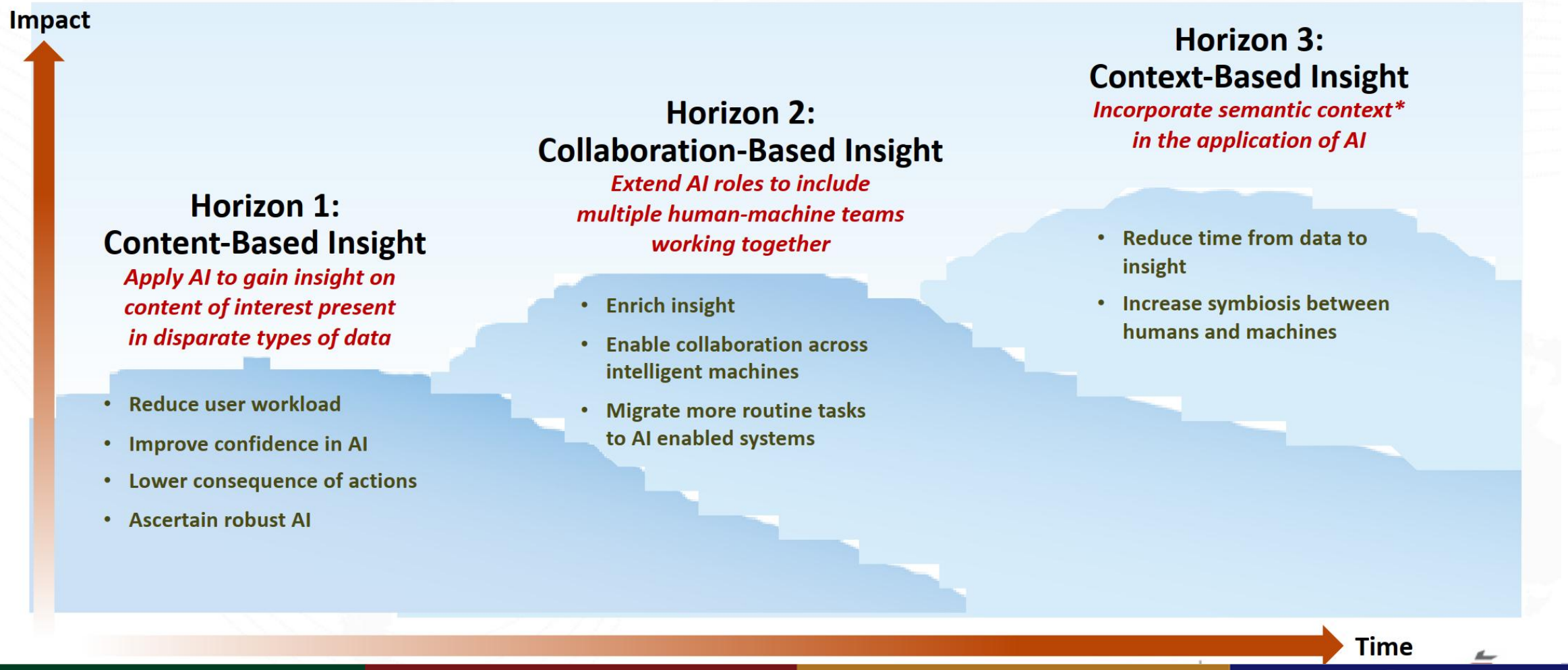
- **Goal:** Resolve customer water-service complaints quickly and accurately
- **Environment:** Customers, water network, pipelines, pumps, valves, field teams, and service systems
- **Percepts:** Customer complaints, account/location data, pressure sensor readings, nearby incidents, and maintenance records
- **Perception:** The agent identifies the issue, affected area, and possible service impact
- **Cognition:** It reasons about likely causes such as leaks, pump failure, closed valves, or high demand
- **Planning:** It decides the next steps, such as checking sensors, reviewing maintenance history, and identifying affected assets

Example: Water Utility Service Response Agent

- **Tool Use:** Accesses GIS maps, SCADA/IoT sensor data, maintenance databases, customer systems, and messaging tools
- **Memory:** Uses past incidents, historical failures, and prior resolutions to improve decisions
- **Self-Reflection:** Checks whether the evidence supports its conclusion and revises its reasoning if needed
- **Action:** Creates work orders, dispatches field teams, updates customers, and escalates major incidents
- **Guardrails:** Protects customer data, limits risky actions, requires approval for critical decisions, and logs all activities

What is an AI agent?

- Three Horizons for Artificial Intelligence Investments



Basic Agent Loop

- An AI agent operates through a **repeated cycle** in which it receives a goal, examines the current situation, selects an appropriate action, observes the result, and decides what to do next.
- Unlike a conventional chatbot that usually produces one response, an agent may perform several controlled steps before completing the task.
- This process is commonly summarized as:

Observe → Understand → Reason → Plan → Use Tools → Act → Reflect → Update Memory → Repeat

- The loop continues until the task is completed, the agent reaches a stopping condition, or human assistance becomes necessary.

Chatbot versus Agent

- **Chatbot interaction**
  - User: What is the status of my reimbursement?
  - Chatbot: Reimbursements usually take several days.
  - Problem:
    - It may invent a general answer.
    - It did not access the actual transaction.
    - It did not consult organizational policy.
- **Agent interaction**
  - User: What is the status of voucher DV-2026-1842?
  - Agent:
    - 1. Calls the reimbursement-status tool.
    - 2. Retrieves the applicable finance procedure.
    - 3. Compares the elapsed processing time.
    - 4. Identifies missing requirements.
    - 5. Generates a grounded response.

Workflow versus Agent

- **Deterministic workflow**
  1. Receive form
  2. Validate fields
  3. Check amount
  4. Apply rule
  5. Send for approval
- Advantages:
  - Predictable
  - Easy to audit
  - Easy to test
  - Appropriate for stable rules

- **Model-directed agent**
  1. Receive message
  2. Interpret intent
  3. Choose relevant tool
  4. Review result
  5. Choose next action
- Advantages:
  - Handles natural language
  - Adapts to different requests
  - Selects tools dynamically
  - Handles incomplete or variable inputs
- Risks:
  - Incorrect tool selection
  - Unsupported decisions
  - Excessive tool calls
  - Prompt injection
  - Unpredictable routing

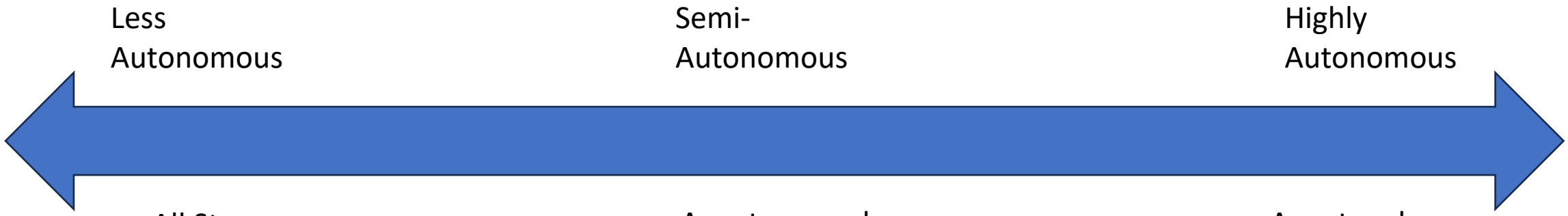
Agent or Not?

- Exercise

1. A model summarizes a meeting transcript.
2. A system searches policy documents and answers with citations.
3. A system classifies a request, retrieves a policy, checks a database, identifies a delegate, and drafts a response.
4. A Python program automatically calculates tax using fixed formulas.
5. A system routes a sensitive case to a human, pauses, and resumes after approval.

Degrees of Autonomy

- Agentic AI can be less or more autonomous



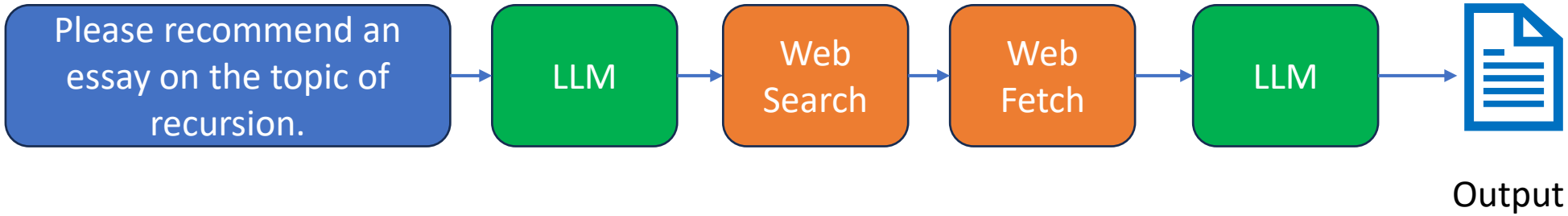
- All Steps Predetermines
- All tool use are hard coded
- Autonomy is just text generation

- Agent can make some decisions and choose tools
- All Tools Predefined

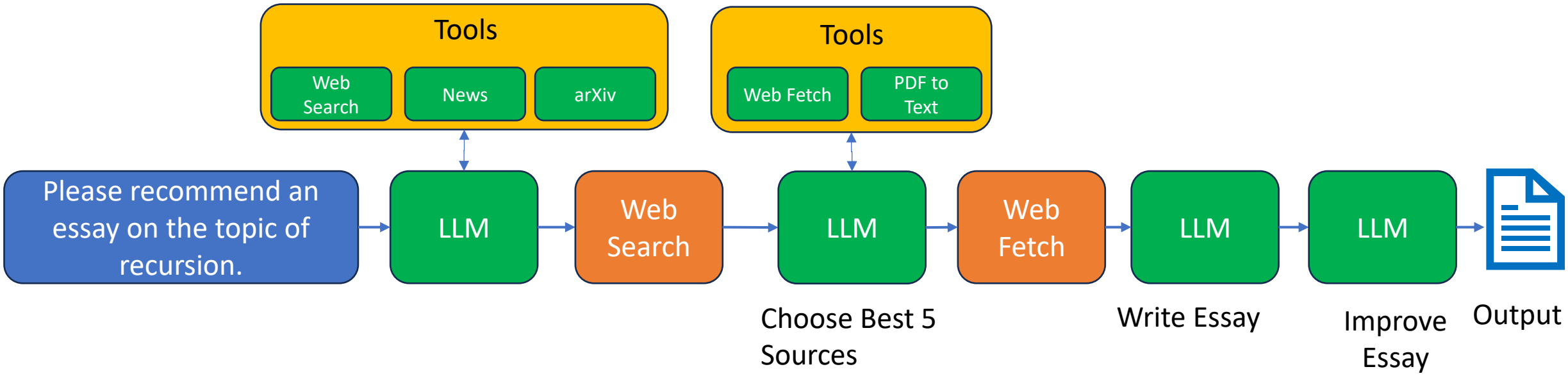
- Agent makes many decisions autonomously
- Can create new tools on the fly

Degrees of Autonomy

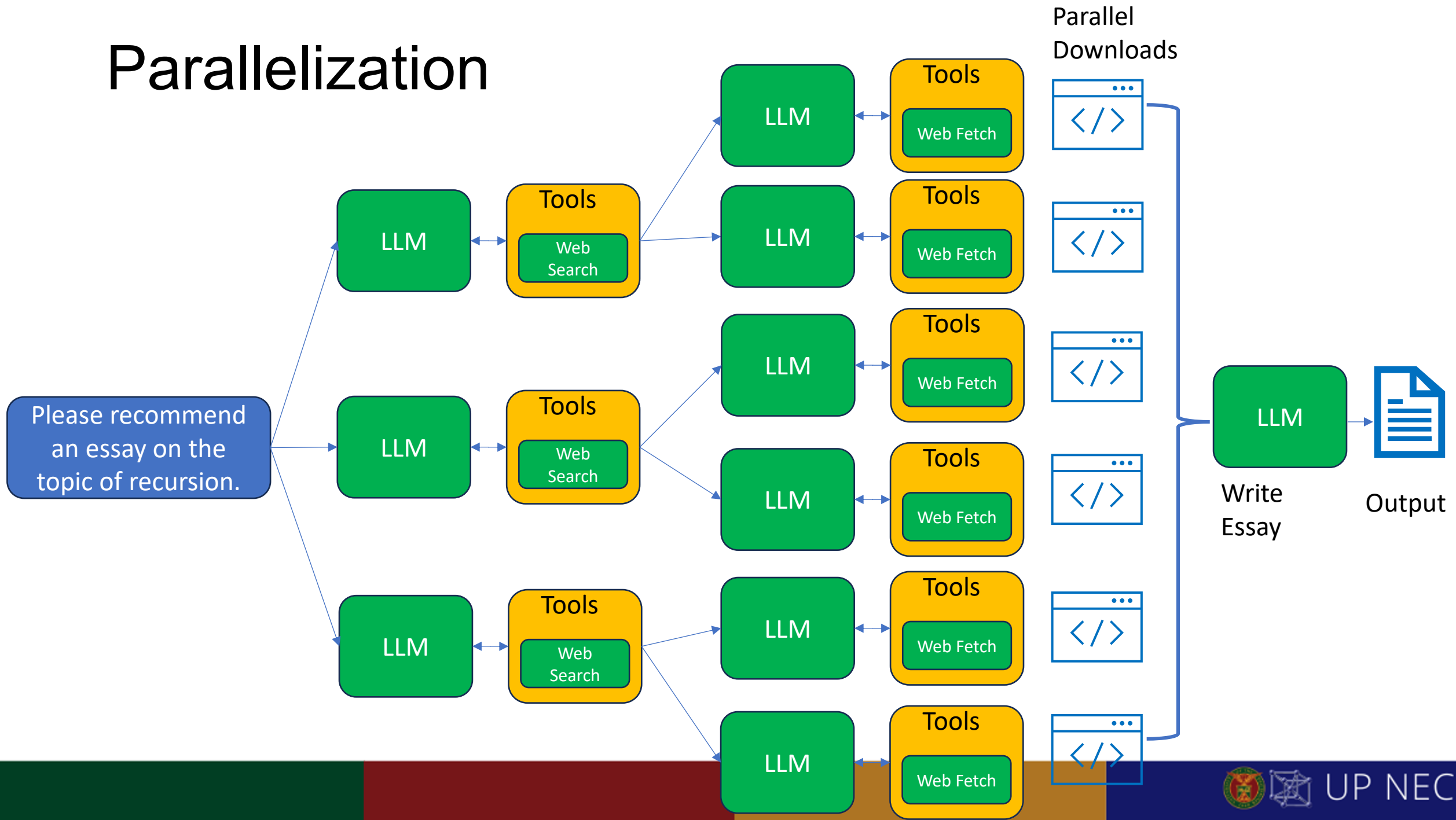
- Less Autonomous



- More Autonomous



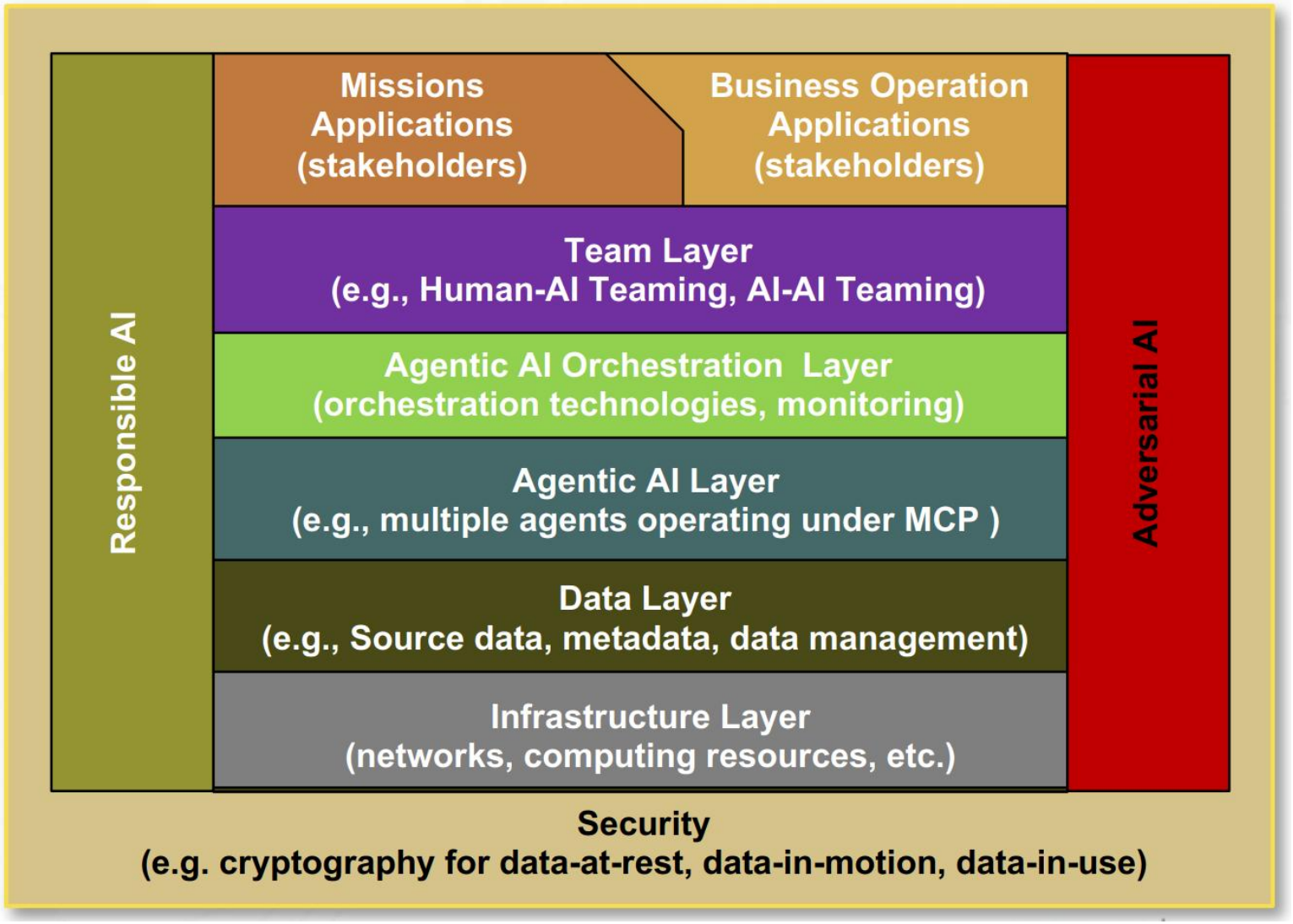
Parallelization



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- **Anatomy of an AI System**
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Anatomy of an AI System



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Mission Applications

- Defines the **problem**, target users, service outcomes, and acceptable impact.
- Converts agent capability into decisions, services, recommendations, or operational support.
- Examples: citizen service triage, research support, emergency response, academic advising, field operations.
- Design question
  - What **mission outcome** improves if agents can reason, retrieve, coordinate, and escalate safely?
- Output:
  - A mission workflow with **measurable value**, human accountability, and defined risk boundaries.

Mission Applications

- **Example: Loan Application Agentic AI Assistant**
  - **Purpose:** Support the bank's lending mission by making loan processing faster, more consistent, and more customer-friendly
  - **Main stakeholders:**
    - Loan applicants
    - Loan officers
    - Bank executives
  - **Mission-level outcomes:**
    - Faster turnaround time for loan applications
    - Better customer experience
    - Higher consistency in loan evaluation
    - Improved compliance with lending policies
    - Better access to financing for qualified borrowers
  - **Example mission goal:**
    - "Approve qualified loans quickly while ensuring large or risky loans receive proper human review"

Business Operation Applications

- Where internal work becomes agent-assisted
- Back-office and internal workflows that improve **organizational productivity and service continuity**.
- Targets repeatable operational processes such as procurement, HR, IT service, finance, and compliance support.
- Uses agents to interpret requests, retrieve procedures, query systems, prepare drafts, and recommend next actions.
- Works best when roles, records, approval points, and service metrics are already defined.
- Design Question:
  - Which operational workflow has enough **volume, structure, and risk controls** to justify an agentic assistant?
- Output
  - A bounded operations use case with clear process owners, tools, and success metrics.

Business Operations

- **Example: Loan Application Agentic AI Assistant**
- **Operational tasks automated or assisted by AI:**
  - Collecting application details
  - Checking document completeness
  - Extracting information from submitted files
  - Validating customer identity
  - Recommending next actions
  - Preparing a summary for human approvers
- **Operational stakeholders:**
  - Frontline branch staff
  - Back-office credit processors
- **Example business rule:**
  - If loan amount ≤ PhP 1,000,000 and all rules pass, AI can recommend approval
  - If loan amount > PhP 1,000,000, AI must escalate to a human approver

Team Layer

- The coordination layer for humans, AI assistants, and specialist agents working together.
- How **people and agents divide work**
  - Defines who initiates, reviews, approves, overrides, and is accountable for the final decision.
  - Supports human-AI teaming, AI-AI teaming, reviewer patterns, supervisors, and specialist handoffs.
  - Turns autonomous behavior into governed collaboration through roles, permissions, and escalation paths.
- Design question
  - Which tasks should be handled by the model, by deterministic code, by a specialist agent, and by a human reviewer?
- Control
  - Role clarity, approval rights, handoff rules, and human oversight before consequential action.

Team Layer

- **Example: Loan Application Agentic AI Assistant**
- **Who is on the team:**
  - Applicant
  - Loan Application AI Assistant
  - Human Approver / Manager
- **How the team works:**
  - AI assists the applicant in submitting requirements
  - AI screens and summarizes the application
  - AI recommends an outcome
  - Human approver reviews applications above PhP 1,000,000
- **Benefit:**
  - Humans focus on high-value judgment tasks while AI handles repetitive work

Agentic AI Orchestration Layer

- The controller that **manages** workflow state, routing, monitoring, and coordination technologies.
- How the system coordinates the agent loop
  - Manages nodes, edges, shared state, conditional routing, retries, checkpoints, and human interrupts.
  - Coordinates extraction, RAG, tools, specialists, approval, draft generation, and final logging.
  - Provides monitoring and execution traces so every step is observable and recoverable.
- Design question
  - What workflow graph should control the sequence from input to decision, approval, response, and audit?
- Mechanism
  - Stateful workflow graph with checkpoints, step limits, error paths, and auditable traces.

Agentic AI Orchestration Layer

- **Example: Loan Application Agentic AI Assistant**
- **What the orchestration layer does:**
  - Starts the workflow when a customer applies
  - Sends the application to the correct specialized agent
  - Monitors whether documents are complete
  - Checks whether escalation is needed
  - Routes loans above PhP 1,000,000 to a human approver
  - Logs every action for audit and compliance
- **Example workflow trigger:**
  - “Application submitted” → orchestration begins intake, validation, scoring, and approval routing

Agentic AI Layer

- The layer where models, agents, tools, and API-connected capabilities **perform bounded work**.
- Where LLMs become action-capable systems
  - Uses local or hosted LLMs to interpret goals, produce structured outputs, select tools, and synthesize observations.
  - Exposes only controlled tools such as policy search, database query, delegate lookup, calculators, and draft generation.
  - APIs can standardize access to external tools, resources, and prompts, but permissions still need application-level controls.
- Design question
  - Which tools should this agent see, and which capabilities must remain unavailable until approval?
- Boundary
  - The model proposes actions; the application validates, authorizes, executes, and logs them.

Agentic AI Layer

- **Example: Loan Application Agentic AI Assistant**
- **Example agents in the solution:**
  - **Intake Agent** — collects applicant information
  - **Document Agent** — reads IDs, payslips, bank statements, and forms
  - **Eligibility Agent** — checks minimum requirements
  - **Credit Assessment Agent** — reviews income, debt, and credit standing
  - **Policy Agent** — checks internal loan rules
  - **Recommendation Agent** — suggests approve / reject / request more documents
  - **Escalation Agent** — routes large loans to a human approver
  - **Customer Communication Agent** — updates the applicant on status
- **Example rule:**
  - If the requested amount is **PhP 1,000,000**, the Escalation Agent sends the case to a human approver

Core Components of the Agentic AI Layer

- **Model**
  - The LLM interprets requests, reasons over context, generates responses, and proposes actions. Usually a pre-trained LLM is used.
- **RAG**
  - Retrieval-Augmented Generation gives the agent access to current organizational knowledge.
- **Tools**
  - Tools allow the agent to query data, calculate values, search documents, create drafts, or trigger workflows.
- **State and Memory**
  - State tracks the current workflow; memory stores useful information across interactions.

The Model as the Agent's Reasoning Engine

- The model is the **language-understanding and reasoning component** of the agent. It interprets goals, understands context, classifies intent, extracts structured information, and generates responses.
- The model can
  - Understand natural-language requests
  - Classify intent and priority
  - Extract structured fields
  - Summarize information
  - Generate recommendations
  - Propose tool calls
  - Draft user-facing responses
- The model should not directly control permissions, execute actions, or make high-risk decisions without validation.

Tools

- Tools are **controlled functions** that allow the agent to do work beyond text generation.
- **Examples**
  - Calculator
  - Date comparison
  - Pandas or CSV query
  - SQLite/database query
  - Policy lookup
  - RAG search
  - Delegate lookup
  - Priority scoring
  - Draft-response generator
  - Ticket creation
- The model may choose a tool, but the application executes the tool only after validation.

Retrieval-Augmented Generation

- RAG allows the agent to answer using **current and approved documents** instead of relying only on the model's internal knowledge.
- **RAG process**
  - User asks a question
  - Query is converted into an embedding
  - Relevant document chunks are retrieved
  - Retrieved context is passed to the model
  - Model generates a grounded answer
  - Citations or source references are validated
- **Use RAG for**
  - Policies
  - Procedures
  - Manuals
  - FAQs

State and Memory in Agentic AI

- Maintains **context** across agent activities
- State and memory allow an agent to retain relevant information rather than treating every model call as an isolated interaction.
  - **State** records the current status of an active workflow.
  - **Memory** preserves selected information for future interactions.
    - Both help the agent maintain continuity, avoid repeated work, and make decisions using prior information.
    - State and memory should be structured, limited, validated, and protected from unauthorized access.
- **Key distinction:**
  State answers, *"What is happening in the current task?"*
  Memory answers, *"What useful information should be remembered later?"*

Data Layer

- The **information foundation** for agents, retrieval, tools, evaluation, and workflow decisions.
- Where evidence, records, and context come from
  - Includes source documents, transaction records, metadata, vector indexes, embeddings, logs, and evaluation datasets.
  - Requires source approval, versioning, access control, data quality checks, and retention rules.
  - Feeds RAG, tool outputs, state updates, metrics, and human-review evidence.
- Design question
  - Which data source is authoritative for each decision, and how will outdated or unapproved sources be excluded?
- Evidence
  - Current, approved, searchable, access-controlled data with metadata and lineage.

Data Layer

- **Example: Loan Application Agentic AI Assistant**
- **Types of data used:**
  - Loan application form data
  - Valid IDs and personal information
  - Proof of income
  - Bank statements
  - Employment details
  - Internal loan policies and approval thresholds
- **Metadata and management needs:**
  - Source of each document
  - Date submitted
  - Version of policies used
  - Confidence score of extracted information
  - Approval history

Example:

The system retrieves the applicant's submitted payslip, income history, and the bank's lending rules before generating a recommendation

Infrastructure Layer

- The **computing and networking foundation** that runs models, tools, storage, and workflows.
- This is where the agentic system actually runs
  - Covers laptops, workstations, DGX Spark, servers, containers, model runtimes, databases, and networks.
  - Supports model serving through local runtimes such as Ollama, API services, vector stores, and workflow persistence.
  - Must be sized for model memory, context length, concurrency, storage, latency, and monitoring.
- Design question
  - What compute, storage, network, and runtime architecture will support the required response time and user volume?
- Platform
  - Reliable model serving, persistent storage, monitoring, backups, and controlled deployment environments.

Infrastructure Layer

- **Example: Loan Application Agentic AI Assistant**
- **Infrastructure components:**
  - Web or mobile loan application portal
  - API connections to bank systems
  - Document storage
  - Databases
  - AI model hosting
  - Workflow/orchestration engine
  - Monitoring dashboard
- **Possible connected enterprise systems:**
  - Core banking system
  - CRM
  - Document management system
  - Notification service (SMS/email)

Example:

Applicant submits documents online, AI services process them in the cloud or private bank servers, and results are sent to internal banking systems

Responsible AI

- A cross-cutting governance layer **that ensures accountability, transparency, and appropriate use**.
- How the system stays accountable
  - Defines intended use, human accountability, review rights, transparency, fairness, privacy, and audit expectations.
  - Requires documentation of model choices, data sources, prompts, schemas, evaluation results, and residual risks.
  - Connects technical controls with organizational governance and stakeholder trust
- Design question
  - Who is accountable for the decision, the tool execution, the data used, and the remaining risk?
- Governance
  - Clear ownership, documentation, reviewer roles, evaluation evidence, and risk acceptance.

Responsible AI

- **Example: Loan Application Agentic AI Assistant**
- **Responsible AI solution:**
  - AI recommendations must be explainable
  - Applicants' personal data must be handled responsibly
  - Loan decisions should not unfairly discriminate
  - High-value loans require human review
  - All decisions and actions must be auditable
- **Example controls:**
  - Provide reasons for rejection or request for additional documents
  - Allow human override of AI recommendations
  - Review model outputs for possible bias

Measure of Performance (MOP)

- A **Measure of Performance** evaluates how well a system, component, agent, or process performs its assigned function. MOPs focus on technical and operational performance that can usually be measured directly during testing or system operation.
- Examples for an Agentic AI system:
  - Classification accuracy
  - Tool-selection accuracy
  - Tool-argument validity
  - Retrieval hit rate at k
- Guiding question:
  - Is the Agentic AI system performing its assigned functions correctly, reliably, and efficiently?

Measure of Performance (MOP)

- **Example: Loan Application Agentic AI Assistant**
  - % of applications screened correctly
  - Average time to check document completeness
  - % of missing documents detected
  - Accuracy of extracted income data
  - % of loans correctly routed to human approver when amount > PhP 1,000,000

Measure of Effectiveness (MOE)

- A **Measure of Effectiveness** evaluates whether the system achieves the intended mission, business, or stakeholder outcome in its actual operating environment. MOEs focus on the value and impact produced by the complete system rather than the performance of an individual component.
- Examples for an Agentic AI system:
  - Reduction in service-request resolution time
  - Reduction in incorrectly routed cases
  - Improvement in policy-compliant decisions
  - Reduction in employee workload
- Guiding question:
  - Is the Agentic AI system helping the organization accomplish its intended mission or business objective?

Measure of Effectiveness (MOE)

- **Example: Loan Application Agentic AI Assistant**
  - Reduction in average loan processing turnaround time
  - % increase in applications completed without rework
  - Reduction in customer waiting time
  - Improvement in customer satisfaction
  - Reduction in loan processing cost per application
  - % reduction in compliance or approval errors
  - Improvement in overall loan officer productivity

MoP and MoE Derivation Process

- A repeatable 4-step methodology for **deriving and instrumenting** MoP and MoE for any new agentic AI system.

MoP Derivation

1. **Define agent-level success criteria**
   What does “correct” mean for each agent’s specific role?
2. **Map to pipeline outcome**
   How do individual agent successes combine into workflow success?
3. **Establish system-level success definition**
   What is the overall mission success condition at the system boundary?
4. **Instrument each level with observable metrics**
   Identify measurable signals at agent, pipeline, and system level — log and alert

MoE Derivation

1. **Map CLEAR dimensions to your use case**
   Set target thresholds for Cost, Latency, Efficacy, Assurance, Reliability
2. **Identify stakeholder value metrics**
   What business outcome must this system improve for each stakeholder group?
3. **Establish pre-AI baseline**
   Record time, cost, and error rates of the process before the agent is deployed
4. **Track continuously in production**
   Monitor CLEAR metrics live; trigger review on drift or threshold breach

Security

- The technical **protection layer** for data, tools, identities, models, and actions.
- How confidentiality, integrity, and control are preserved
  - Protects data at rest, in motion, and in use through encryption, identity, access control, and secrets management.
  - Enforces tool allowlists, role-based access, input validation, output redaction, and approval before write actions.
  - Treats user input, retrieved documents, tool results, and agent messages as untrusted until validated.
- Design Question:
  - Which data and tools require stronger controls before the model can access, use, or act on them?
- Protection
  - Least privilege, read-only defaults, approved recipients, encrypted storage, and safe logging.

Security

- **Example: Loan Application Agentic AI Assistant**
- **Security focus areas:**
  - Protect applicant personal and financial data
  - Prevent unauthorized access to loan records
  - Secure communications between AI agents and enterprise systems
  - Protect data in use during AI processing
- **Example:**
  - Only authorized credit officers can view full loan applications, and all actions are recorded for audit purposes

Adversarial AI

- The threat layer that **tests** how the system behaves under manipulation, poisoning, and misuse.
- How the system is attacked and stress-tested
  - Covers direct prompt injection, indirect injection, jailbreaking, data poisoning, phishing, and tool misuse.
  - Tests whether the system blocks unsafe tools, unapproved recipients, sensitive disclosure, and goal hijacking.
  - Uses red-team cases and regression tests to strengthen guardrails and measure residual risk.
- Design Question
  - What adversarial inputs could cause the agent to select the wrong tool, leak data, or bypass human approval?
- Assurance
  - Security test set, attack simulations, before-and-after guardrail evaluation, and risk register.

Adversarial AI

- **Example: Loan Application Agentic AI Assistant**
- **Examples of adversarial or malicious risks:**
  - Fake payslips or altered bank statements
  - Identity fraud
  - Prompt injection against AI agents
  - Attempts to manipulate eligibility checks
- **How the system should respond:**
  - Detect suspicious document patterns
  - Cross-check data across sources
  - Flag low-confidence extractions
  - Escalate suspicious cases for human investigation
  - Block unsafe prompts or unauthorized actions
- **Example:**
  - If the AI detects inconsistencies between declared salary and submitted bank statements, it flags the application for fraud review

What Are Guardrails?

- Guardrails keep agent behavior **safe** and **controlled**
- Guardrails are controls that prevent, detect, redirect, block, or record unsafe or unreliable agent behavior.
- They protect the system from:
  - Prompt injection
  - Unauthorized tools
  - Invalid tool arguments
  - Sensitive-data exposure
  - Incorrect routing
  - Unsupported answers
  - Excessive autonomy
  - Repeated tool calls
  - Unsafe external actions
- Guardrails must operate across the entire system—not only inside the prompt.

Design Checklist for an AI system

- 10 Guidelines to Successfully Deploy AI System Capabilities
  - 1. Scope the Agentic AI application meeting the **customer needs**
  - 2. Develop an end-to-end Agentic AI pipeline anchored on an AI system architecture
  - 3. Demonstrate an initial Agentic AI capability then iterate adhering to an LLMOps methodology
  - 4. Verify individual subcomponents and validate end-to-end Agentic AI system
  - 5. Secure Agentic AI system both physically and against cyber threats
  - 6. Attend to Responsible AI and adversarial vulnerabilities
  - 7. Strengthen AI team by fostering internal and external relationships
  - 8. Build a multidisciplinary and diverse team with complementary skills
  - 9. Provide measurable objectives while mentoring AI talent
  - 10. Continue to expand AI team skills as future of work evolves

Outline for this Session

- Introduction to AI
- Agentic AI
- Anatomy of an AI System
- **Project Feasibility and Risk Management**
- Selecting AI Applications
- Project Proposal Template

Project Feasibility and Risk

- Feasibility Has Several Dimensions
- Evaluate:
  - Process feasibility
  - Data feasibility
  - Model feasibility
  - Integration feasibility
  - Infrastructure feasibility
  - Financial feasibility
  - Organizational feasibility
  - Security and compliance feasibility
- A technically possible project may still be organizationally infeasible.

Process Feasibility

- Ask:
  - Is the current process documented?
  - Are decision rules understood?
  - Are responsibilities clear?
  - Are exceptions known?
  - Is there an agreed escalation process?
  - Is the proposed agent solving the correct problem?
- Avoid automating a **broken or undefined** process.

Data and Knowledge Readiness

- Can the agent **access** authoritative information?
- Evaluate:
  - Availability
  - Quality
  - Completeness
  - Consistency
  - Timeliness
  - Accessibility
  - Ownership
  - Confidentiality
  - Format
  - Version control

Model Capability Feasibility

- Ask whether the selected model can **perform the task reliably** enough under actual operating conditions.
- Determine whether the required task involves:
  - Classification
  - Extraction
  - Summarization
  - Retrieval
  - Generation
  - Reasoning
  - Planning
  - Tool selection
  - Multimodal processing

Integration Feasibility

- A strong model cannot compensate for **unavailable integrations.**
- Consider:
  - Availability of APIs
  - Database access
  - Authentication
  - Permission scopes
  - Network restrictions
  - Legacy systems
  - Tool response formats
  - Rate limits
  - Sandbox or testing environments
  - Write-access controls

Infrastructure and Performance

- Do we have enough **computing** power?
- Assess:
  - Local versus cloud deployment
  - CPU, GPU, and memory requirements
  - Model size
  - Expected transaction volume
  - Response-time requirement
  - Availability requirement
  - Storage
  - Backup and recovery
  - Scalability
  - Connectivity

Privacy and Confidentiality

- Questions to address:
  - What personal information is processed?
  - What confidential organizational information is involved?
  - Can data be sent to an external model?
  - How long will inputs and outputs be stored?
  - Who can access execution logs?
  - Should information be anonymized?
  - Can the model expose information from another user?

Security Risks

- Important risks include:
  - Prompt injection
  - Indirect prompt injection
  - Malicious documents
  - Unauthorized tool use
  - Credential exposure
  - Excessive permissions
  - Data leakage
  - Impersonation
  - Phishing
  - Manipulated links
  - Malicious attachments
  - Unauthorized routing

Operational Risks

- Possible operational failures:
  - Incorrect classification
  - Missed urgent request
  - Incorrect recipient
  - Unsupported recommendation
  - Failed tool call
  - Duplicate action
  - Infinite agent loop
  - Outdated policy retrieval
  - Incomplete audit record
  - Failure to escalate
  - Service interruption

Human Oversight Design

- Define:
  - Which actions require approval
  - Who may approve them
  - How approval is recorded
  - What happens when no approver is available
  - How the user can correct the agent
  - How the agent communicates uncertainty
  - When the agent must abstain
  - How overrides are reviewed

Cost and Benefit Feasibility

- Potential costs:
  - Model inference
  - Computing hardware
  - Software development
  - Integration
  - Data preparation
  - Security controls
  - Evaluation
  - Monitoring
  - User training
  - Maintenance
- Potential benefits:
  - Time savings
  - Reduced backlog
  - Reduced rework
  - Better decisions
  - Faster service
  - Improved continuity

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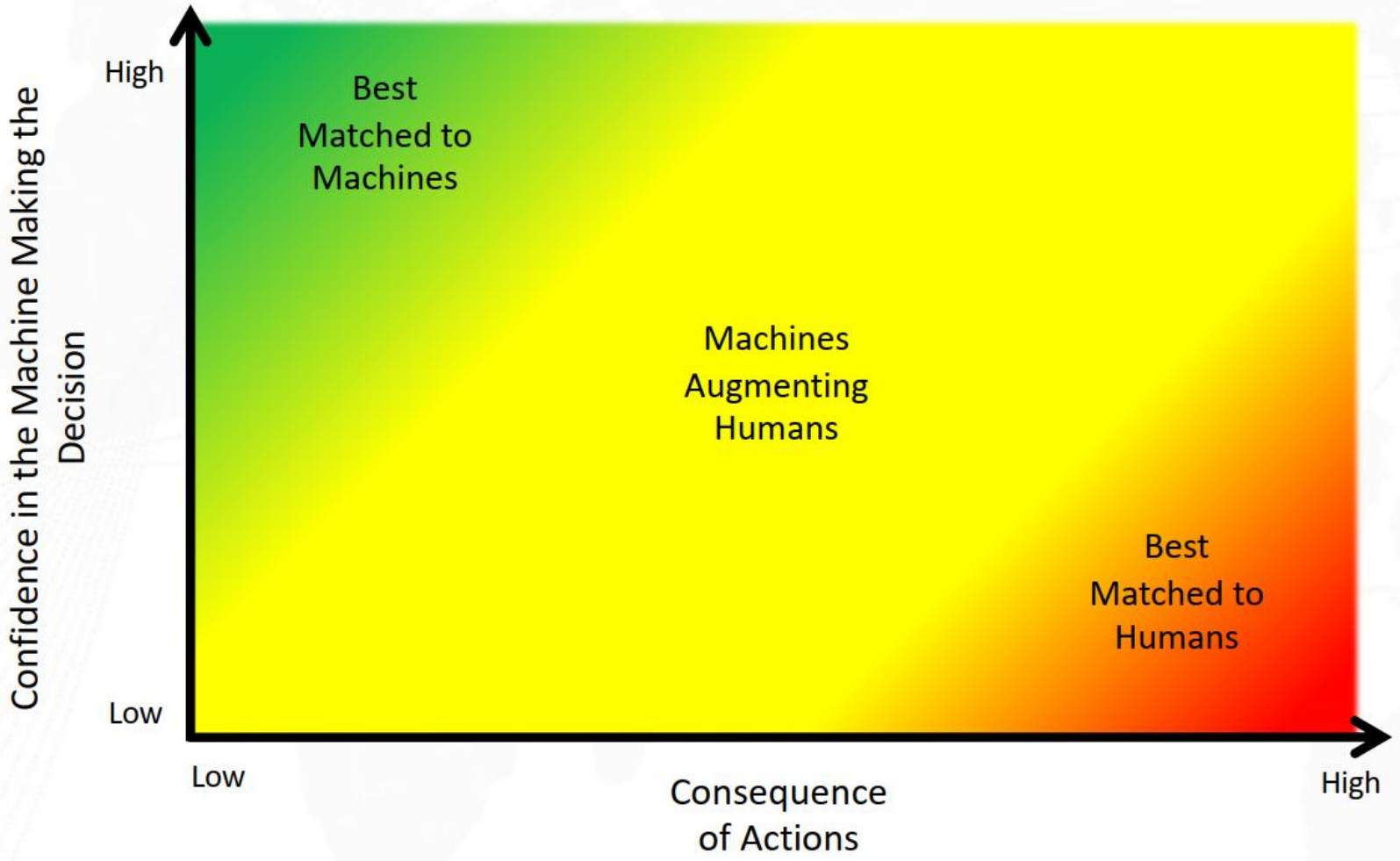
Selecting Appropriate AI Applications

- Characteristics of a **good AI project**
- A suitable project usually has:
  - A clear operational goal
  - Repetitive information-processing work
  - Multiple data or document sources
  - Decisions requiring language interpretation
  - Clearly identifiable tools
  - Defined escalation rules
  - Measurable outputs
  - Available human reviewers
  - Controllable risk
  - Accessible test data

Poor candidate applications

- AI may be **inappropriate** when:
  - The task is fully defined by stable rules.
  - Errors could cause immediate irreversible harm.
  - No reliable source data exist.
  - No human owner is accountable.
  - The required systems cannot be safely integrated.
  - The action cannot be audited.
  - The organization has no process for handling failures.
  - A simple search, form, or script would solve the problem.
  - The project has no measurable success criteria.

Confidence Level vs. Courses of Action



AI suitability matrix

Dimension	Low suitability	High suitability
Language variability	Standard fields	Free-form messages
Tool requirement	No external data	Multiple controlled systems
Process variability	Fixed sequence	Conditional routing
Risk	Irreversible/high stakes	Bounded and reviewable
Evaluation	No clear success measure	Labeled cases and metrics

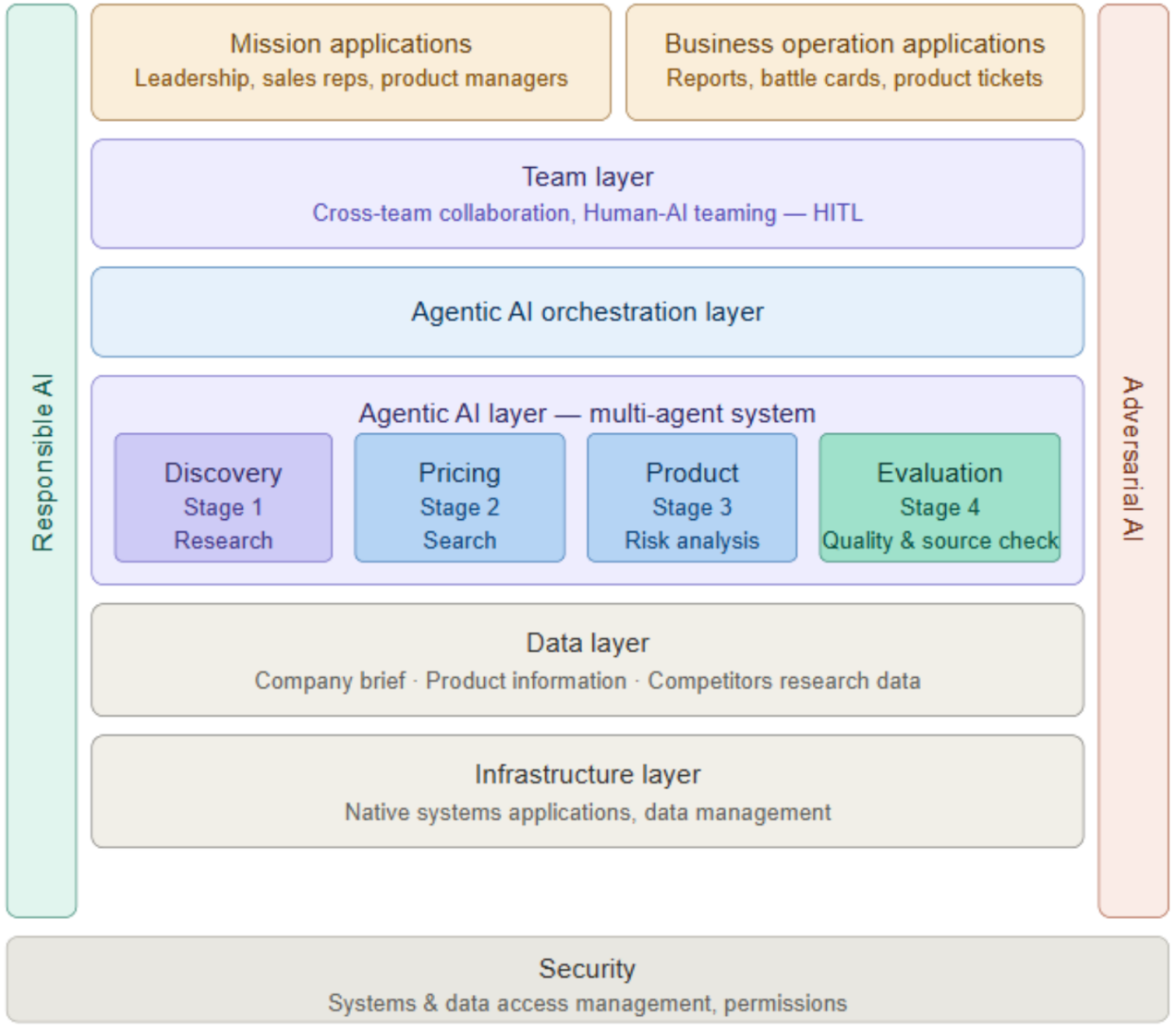
Business Value/Drivers

Key Question	Your Answer
What insight is the customer needing?	*[Explain why Agentic AI would be useful for this customer]*
What Use Case is most relevant to your Agentic AI application?	*[Name the specific business problem being solved]*
Why is this Use Case most appropriate for your team application?	*[Make or OpenAI use case. Is your solution revolutionary or evolutionary vs. current approaches?]*
Why is Agentic AI the right solution?	*[Justify the specific benefit of employing Agentic AI here]*
What data input is required to achieve the desired insight?	*[List prompting strategies and domain-specific data needed]*
How would you address Responsible AI?	*[Explain the most important guardrails relevant to your application]*
How would you determine performance of your Agentic AI system?	*[Describe MoP: analytical metrics (Precision, Recall, F1) + MoE: business impact]*

Agentic AI Use Case: Overview

Field	Your Content (replace this text)
Use Case Name	*[Short descriptive name, e.g. Pharma Customer Complaint Router]*
Business Domain	*[Industry and function, e.g. Pharmaceuticals — Quality & Compliance]*
Problem Statement	*[One sentence: what manual or broken process does this replace?]*
Agentic AI Approach	*[Agent pattern: single / multi-agent? Router? RAG? Tool-use?]*
Input Data	*[All input sources: email, CRM, ERP, documents, APIs, etc.]*
Output / Action	*[What does the agent produce or trigger? Be specific.]*
Platform & Tools	*[OpenAI Agent Builder / Make.com / other]*
Key Prompting Strategy	*[Few-shot? CoT? Structured output? Domain-specific system prompt?]*
Guardrails	*[Confidence threshold, HITL trigger conditions, fallback logic]*
MoP — Performance	*[Analytical metrics: Precision, Recall, F1-Score, Latency, etc.]*
MoE — Effectiveness	*[Business impact: ticket resolution time, throughput, cost savings]*

Use Case Example Agentic AI Architecture



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Enterprises already using Agentic AI

- Software development: AI agents that can write and debug code independently
- Customer service: Advanced chatbots that can resolve complex issues without human assistance
- Supply chain: AI agents that manage inventory, forecast demand and optimize delivery routes
- Manufacturing: Systems that monitor equipment detect defects and streamline production lines
- Cybersecurity: Agents that detect threats, isolate risks and respond in real time
- Insurance: AI agents that process claims, assess risk and draft responses
- Banking: Systems that verify income, credit and approve loans
- Healthcare: Tools that flag anomalies, suggest diagnoses and draft notes

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Case 1: Project Proposal Template

- Teams discuss and select one Agentic AI application for the project suitable for implementation given the risks and feasibility of available data.
- Each team identifies a real organizational process that could benefit from an Agentic AI solution. Describes the business problem, target users, expected benefits, available data, and what the proposed agent will be allowed to do.
- Answer the next four slides:
  - Business Value/Drivers
  - Agentic AI Use Case
  - Agentic AI Architecture
  - MoP and MoE
- Proposals will be presented before lunch.

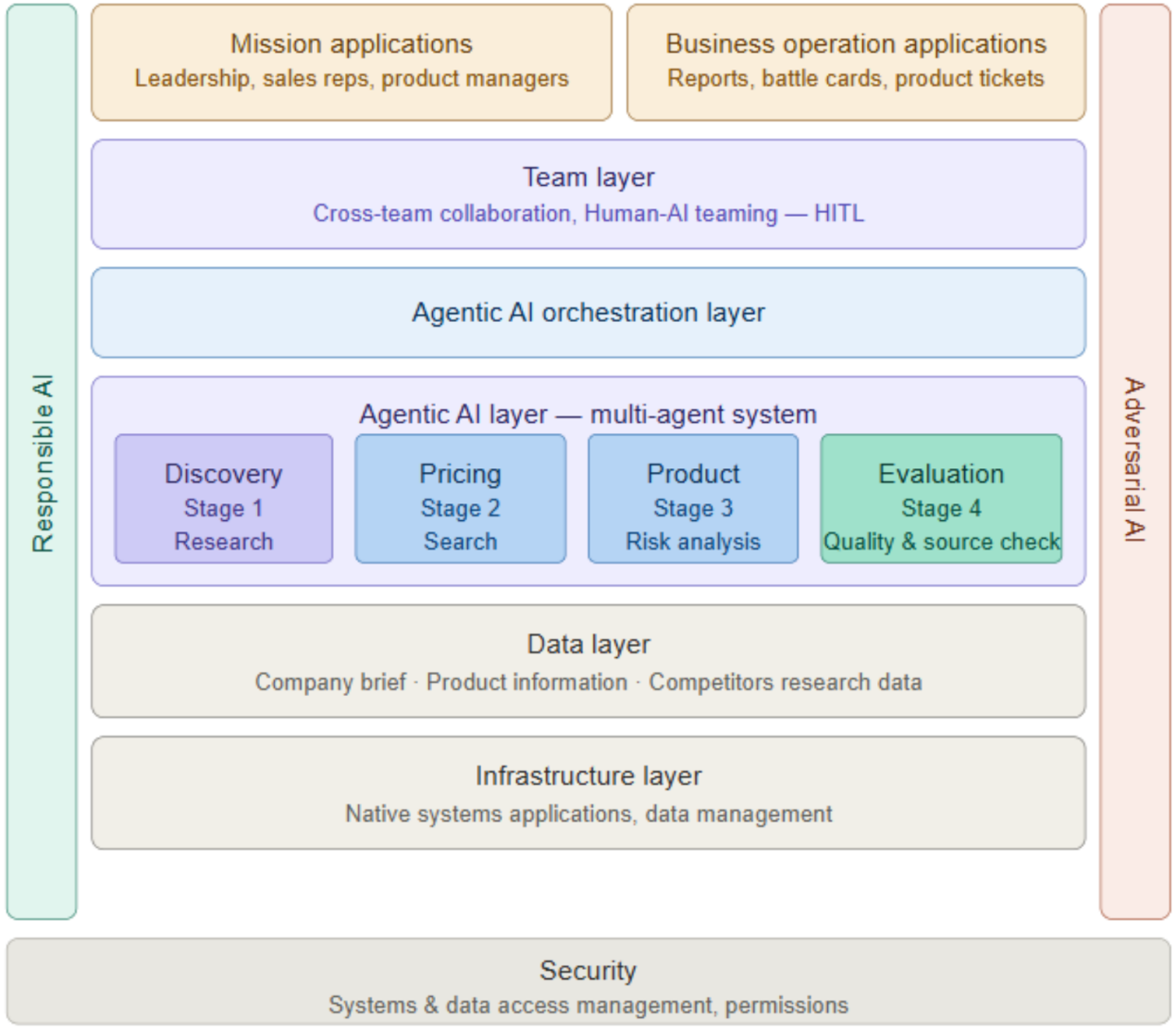
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Agentic AI Architecture



MoP and MoE

- Use the MoP and MoE Derivation Process to derive your proposed project's Measures of Performance and Measures of Effectiveness

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References

- David Martinez; Artificial Intelligence: A Systems Approach from Architecture Principles to Deployment (MIT Lincoln Laboratory Series) 2025
- Andrew Ng (Deep Learning AI)